Touch is Vision

Qing Gu

This collection showcases my work in creating accessible solutions
for the needs of visually impaired children,
specifically focusing on areas such as inclusive transportation,
geometry education, physics education,
and low-cost modifications of accessibility facilities.

Content

#1

Myway

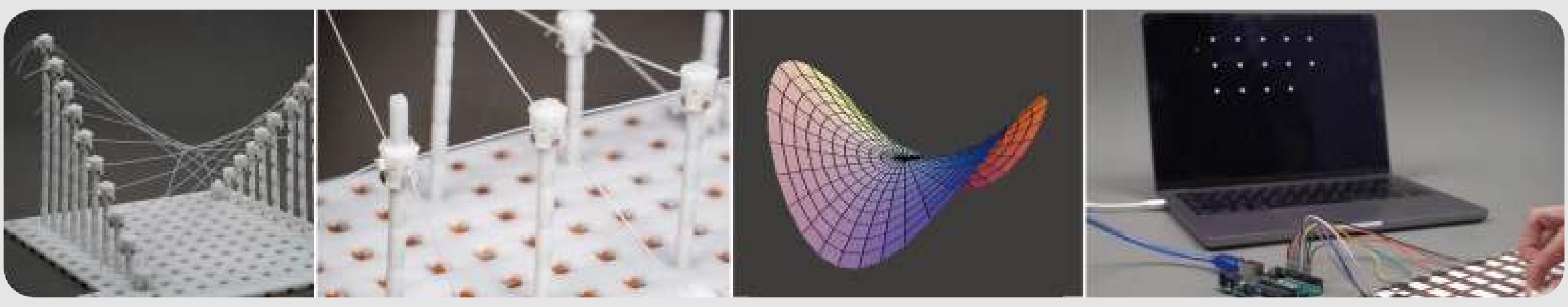
Myway is a 3D modular kit with Braille, tactile paving, and spatial audio to facilitate visually impaired teenagers in comprehending public transportation scenarios. The research with 25 visually impaired users from the school for the blind highlighted its effectiveness in enhancing comprehension and promoting independent travel confidence.
Manuscript under review at CHI2024.

*Collaborate with Xiang Li

#2

TactiGeo

TactiGeo is a modular and scalable tangible interaction system that constructs wireframe representations of geometric entities. This system allows visually impaired users to explore three-dimensional geometric models through touch. Multiple rounds of usability testing and semi-structured interviews with both experts and students have validated the effectiveness of TactiGeo in facilitating the understanding of spatial geometry knowledge.

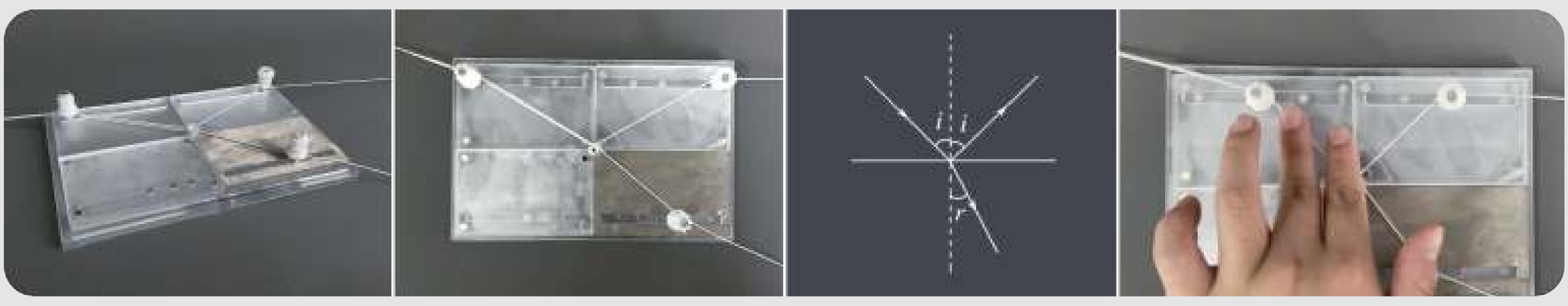


*Collaborate with Xiang Li

#3

TactiLight

TactiLight is a modular and cost-effective tangible interaction system with various tactile feedback and auditory guidance. It allows users to explore concepts such as light reflection, and refraction by manipulating different incident materials and selecting various angles of incidence. Based on simple tests conducted with experts and visually impaired students, TactiLight has proven to be effective in aiding their understanding of optical concepts.



#4

BrailleSticker

BrailleSticker is a 3D-printed sticker designed for low-cost modifications of accessibility devices. Visually impaired individuals can submit printing requests to designers for items, such as a touchscreen password lock, which can be parametrically printed. They can subsequently affix these Braille stickers to a wide range of objects within the environment, thereby facilitating enhanced object identification and utilization.

#1 Myway

Myway is a 3D modular kit with Braille, tactile paving, and spatial audio to facilitate visually impaired teenagersin comprehending public transportation scenarios. The research with 25 visually impaired users from the school for the blind highlighted its effectivenessin enhancing comprehension and promoting independent travel confidence.

Published: Gu, Q., Li, X., Cai, Q. et al. MyWay: a 3D and audio-enhanced transportation learning kit for the visually impaired teenagers. CCF Trans. Pervasive Comp. Interact. 7, 151–173 (2025). https://doi.org/10.1007/s42486-024-00163-y

*Collaborate with Xiang Li

Background Research

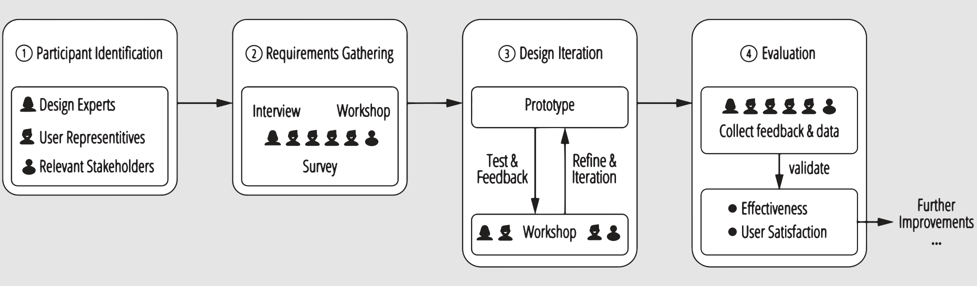
Requirements

Interviews with participants and co-design workshop helped us identified two **key features** of the learning kit:

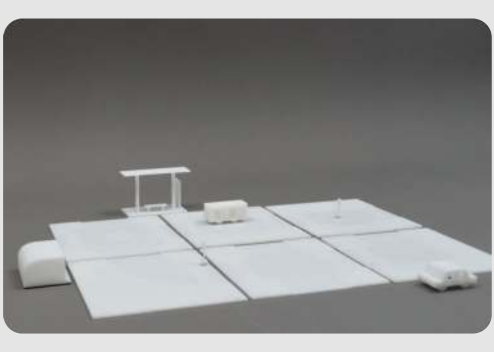
1. Fundamental orientation guidance: Tactile paving is essential for users to navigate their movement trajectory and direction. The learning kit should therefore include modules describing **tactile paving**, especially at critical points like t**urns, intersections, and entrances**. Additionally, **facility modules** such as escalators, elevators, accessible entryways, buses, and subway cars are crucial for orientation and should be incorporated.

2. Realistic spatial experience with diverse feedback: Beyond **static tactile cues** for orientation, **sound is also a vital component**, necessitating the provision of a rich, **multi-layered**, and **realistic auditory experience**.

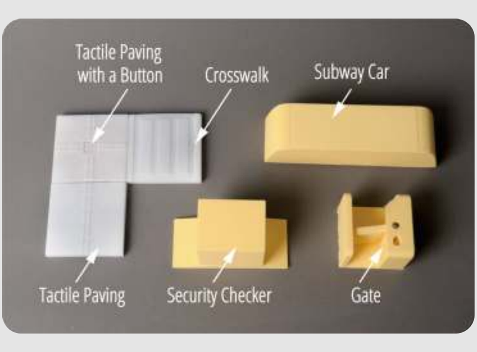
Design Refining



The entire co-design process

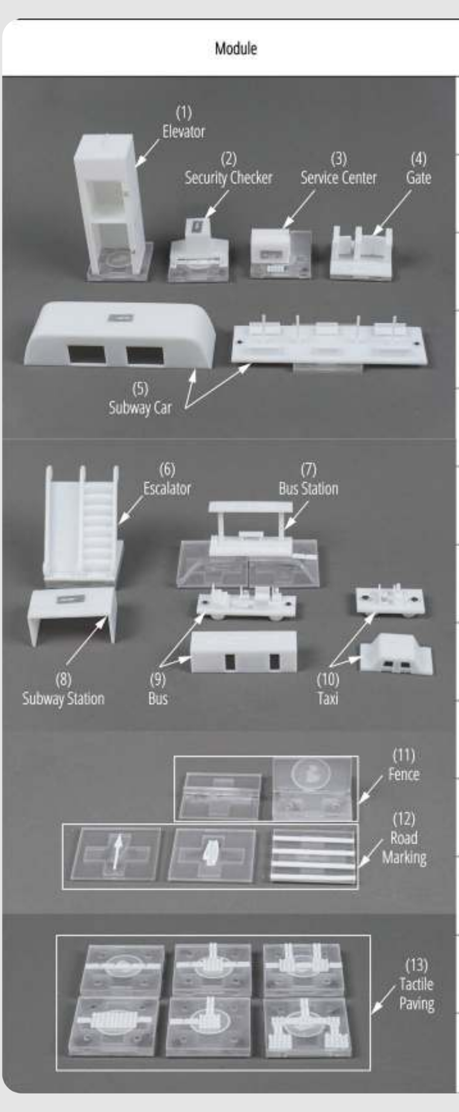


The first test version



The second test version

Through user testing and evaluation of the prototype with these 9 users, we identified several key features of the learning kit:

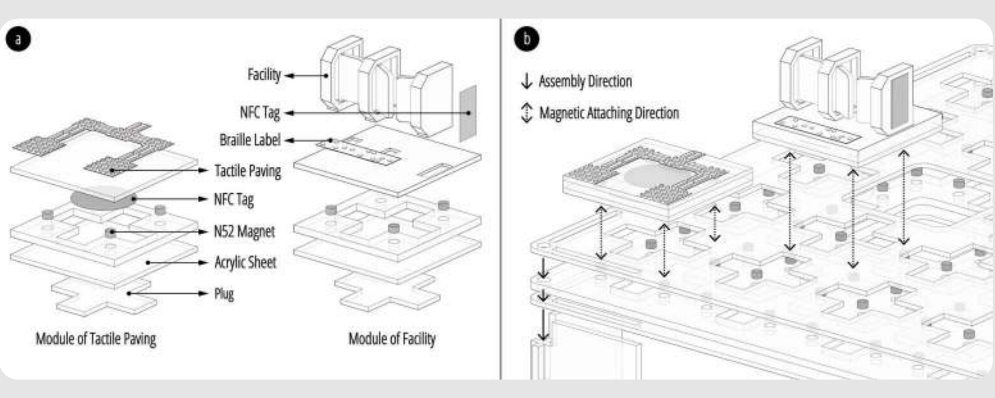


Name	Original Facility	Tactile Feedback	Audio Feedback
(1) Elevator		Interactive Button Movable carriage Tactile Paving & Braille	Explanation
(2) Security Checker		Tactile Paving & Braille	Info - Broadcast Env - Noise Explanation
(3) Service Center		Tactile Paving & Braille	Explanation
(4) Gate		Movable Door Panels Tactile Paving & Braille	Info - Beep Explanation
(5) Subway Car		Openable Shell Braille	Info - Broadcast Env - Noise Explanation
(6) Escalator		Tactile Paving	Info - Broadcast Env - Noise Explanation
(7) Bus Station		Tactile Paving & Braille	Explanation
(8) Subway Station		Tactile Paving	Explanation
(9) Bus		Openable Shell Braille	Info - Broadcast Env - Noise Explanation
(10) Taxi		Openable Shell Braille	Info - Honk Env - Noise Explanation
(11) Fence		Abstract Static Entity	\
(12) Road Marking		Tactile Graphics	\
(13) Tactile Paving		Tactile Paving	\

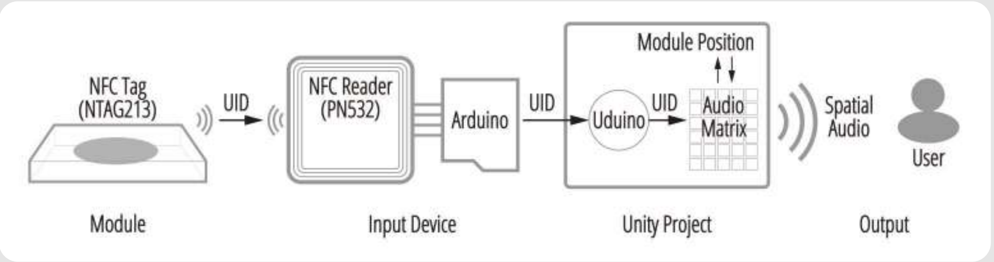
System Overview

Watch the video → https://youtu.be/iUnqmPxARBc

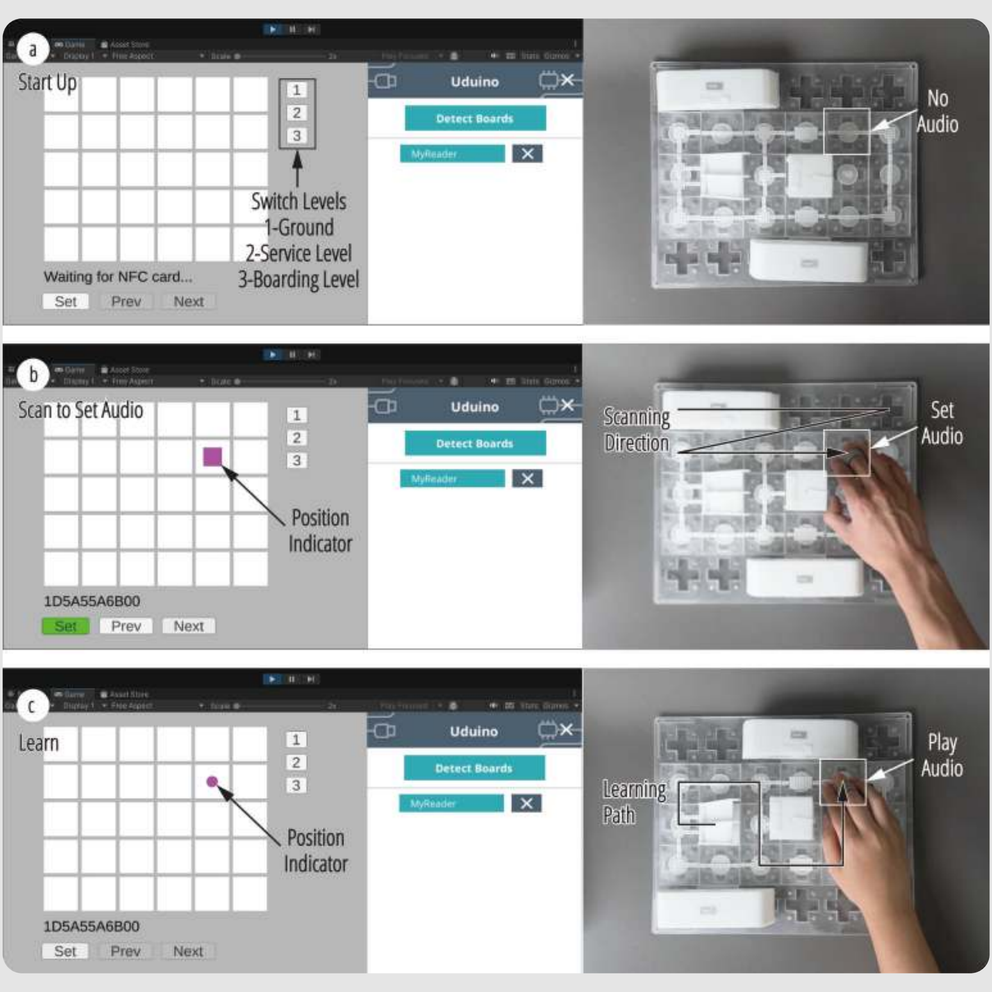
The kit is divided into three levels: (a) Ground Level, (b) Concourse Level, and (c) Platform Level.



Manufacture and structure: (a) Composition of manufactured parts for two typical modules. (b) The structure of the baseboard and how the modules are placed on the baseboard.



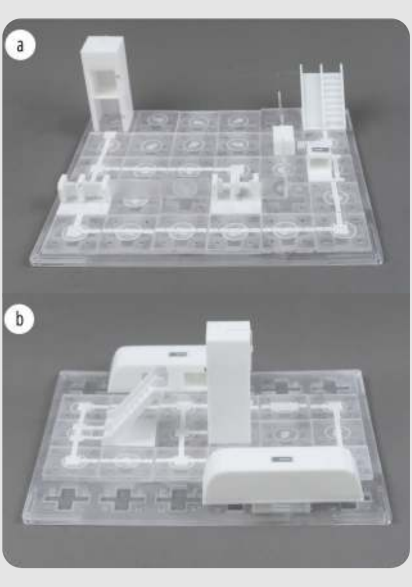
Technical flow for reading NFCtag data and final output of spatial audio.



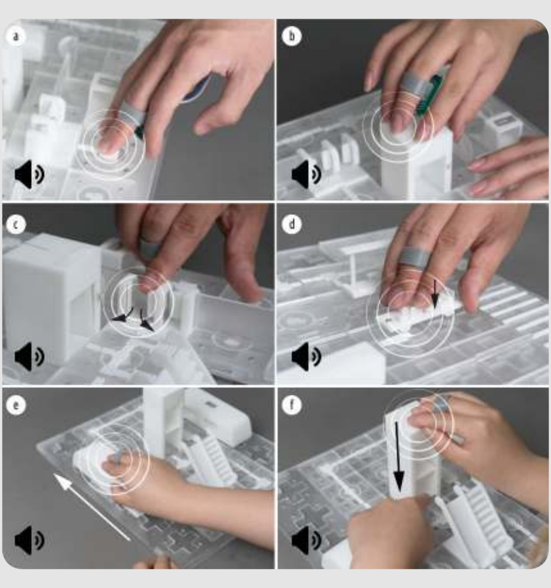
Behavior and corresponding interface: (a) Startup: The left side displays information and operational buttons, while the right side detects connected card reader hardware, (b) Sighted partner scanning: The corresponding NFCtag ID is assigned to the module, and (c) Visually impaired learner's touch interaction: User interface indicates the module currently being touched by the user.

How to Use

The construction process involves the following steps: (a) Determining module placement, (b) Constructing roads, (c) Setting up additional auxiliary facilities, and prepare to scan modules and generate spatial audio.



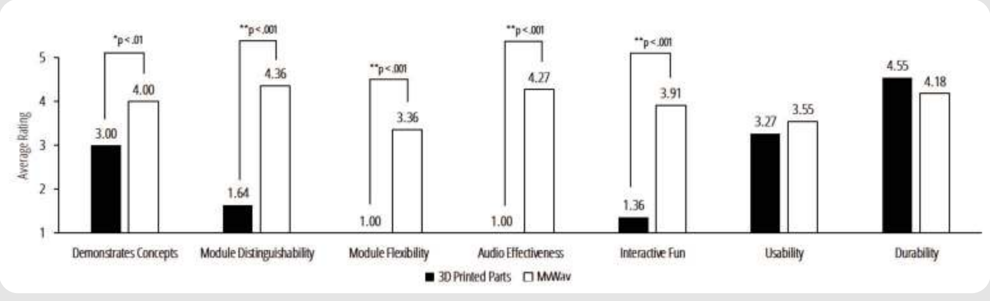
The modular structure allows for customized assembly: (a) the original simplest route, and (b) a customized, more complex route.



The usage process of the visually impaired user: Touch Road, understand Module, move Module, listen to audio.

User Study

In the culminating phase of user testing, **we recruited a group of 10 teenagers from a schoolfor the blind** (M_age = 16.70; 6 males, 4 females). Additionally, our experimental group consisted of **one expert in the field of O&M** (M_age = 30.00; 1 female), along with 5 sighted individuals (M_age = 22.80; 3 males, 2 females) who actively participated as tutors during the instructional

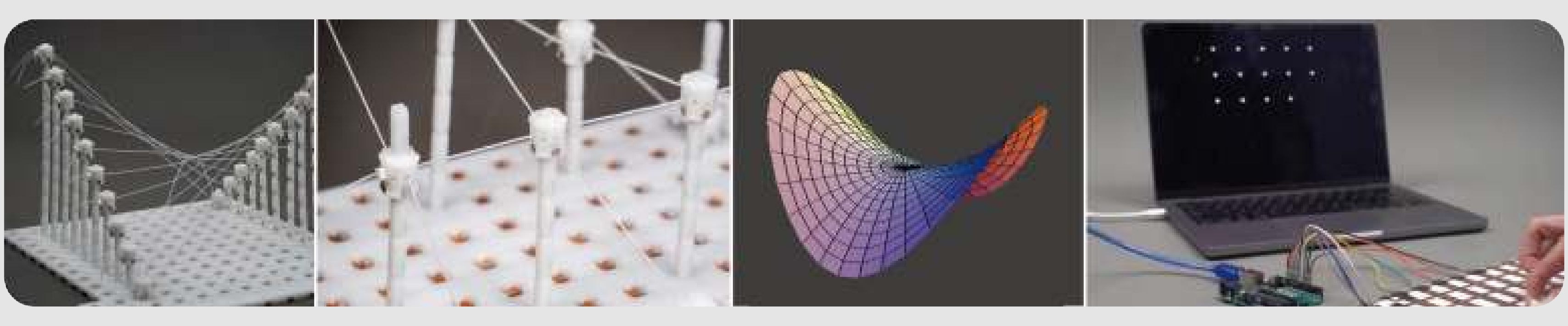


Participants rated each indicator for the use of traditional 3D printed parts and Myway. ** indicates a highly significant difference ($p < .001$), * indicates a significant difference ($p < .01$), **suggesting a significant difference in usage between traditional 3D printed parts and Myway**.

The 16 participants all completed a survey (Five-point Likert Scale). They expressed high appreciation for the modules' ability to **demonstrate concepts** (M = 4.19, SD = 0.63). They also found the kit to be highly distinguishable, effectively identifying and understanding each module's function (**Module Distinguishability** M = 4.38, SD = 0.48). Furthermore, they perceived the audio inclusion as a significant enhancement to the kit's **effectiveness** (M = 4.13, SD = 0.70). In terms of **durability**, they rated the kit highly (M = 4.19, SD = 0.73) and also found it **interactively engaging** (M = 3.94, SD = 0.56). Overall, the participants were satisfied with the kit's **flexibility** (M = 3.60, SD = 0.70) and usability (M = 3.80, SD = 0.73). However, some of the blind school students did highlight certain learning barriers.

#2 TactiGeo

TactiGeo is a modular and scalable tangible interaction system that constructs wireframe representations of geometric entities. This system allows visually impaired users to explore three-dimensional geometric models through touch. Multiple rounds of usability testing and semi-structured interviews with both experts and students have validated the effectiveness of TactiGeo in facilitating the understanding of spatial geometry knowledge.



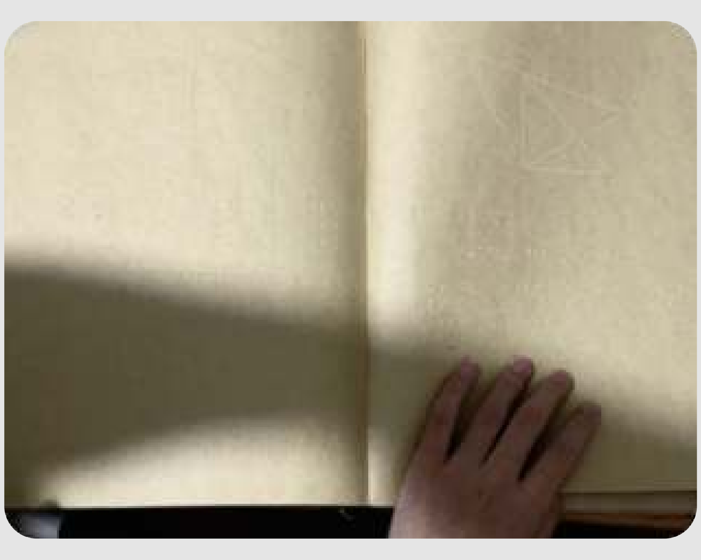
*Collaborate with Xiang Li

Background Research

In the absence of adequate **teaching aids** and with **limited instructional methods**, teachers find it challenging to **convey spatial information** to students solely through limited dimensions of **oral descriptions and Braille information**.
Additionally, due to their lack of visual experience, students struggle to comprehend **abstract geometric concepts** and often **lack motivation and willingness** to learn.

Limited oral descriptions

Limited teaching aids and methods

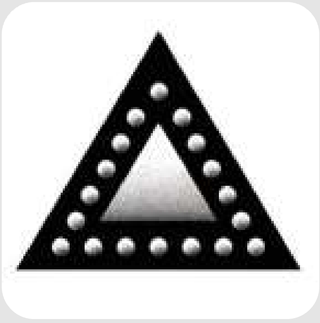


Tactile Graphics Made of Braille Dot Patterns

Requirements

From semi-structured interviews with visually impaired students and blind school teachers, we identified **several requirements**:

1.Describe abstract complex concepts



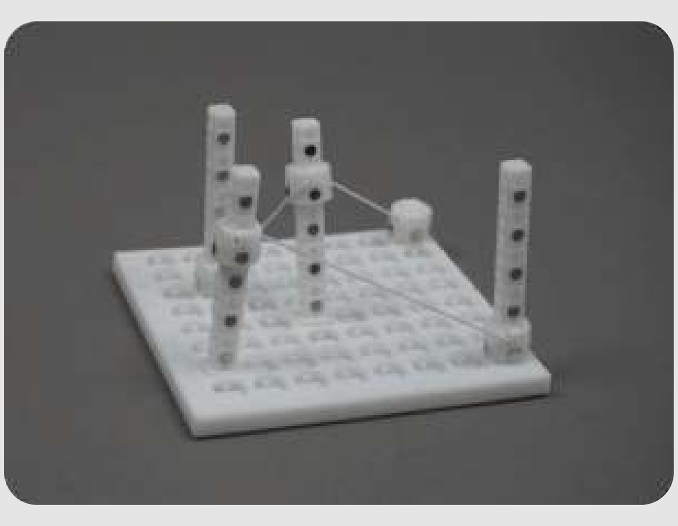
2.Improve recognition of Braille dot

3.easily and freely assembled

4.Sound-assisted learning

Design Refining

Initial test version



The second test version

Through user testing and evaluation of the prototype with **6 users**, we identified the design concept and the required elements of the system:

1.Enhancing tactile display

Improving the rigidity of **wireframe structures,** reducing **module size**, and enhancing the **stability of connections** to minimize the impact of physical structures on the tactile presentation of geometric models.

2.Improving usability for visually impaired users

Designing **simpler structures** for visually impaired users, such as **magnetic connections** or sliding mechanisms, and providing additional information like Braille annotations.

3.Enhancing assembly efficiency

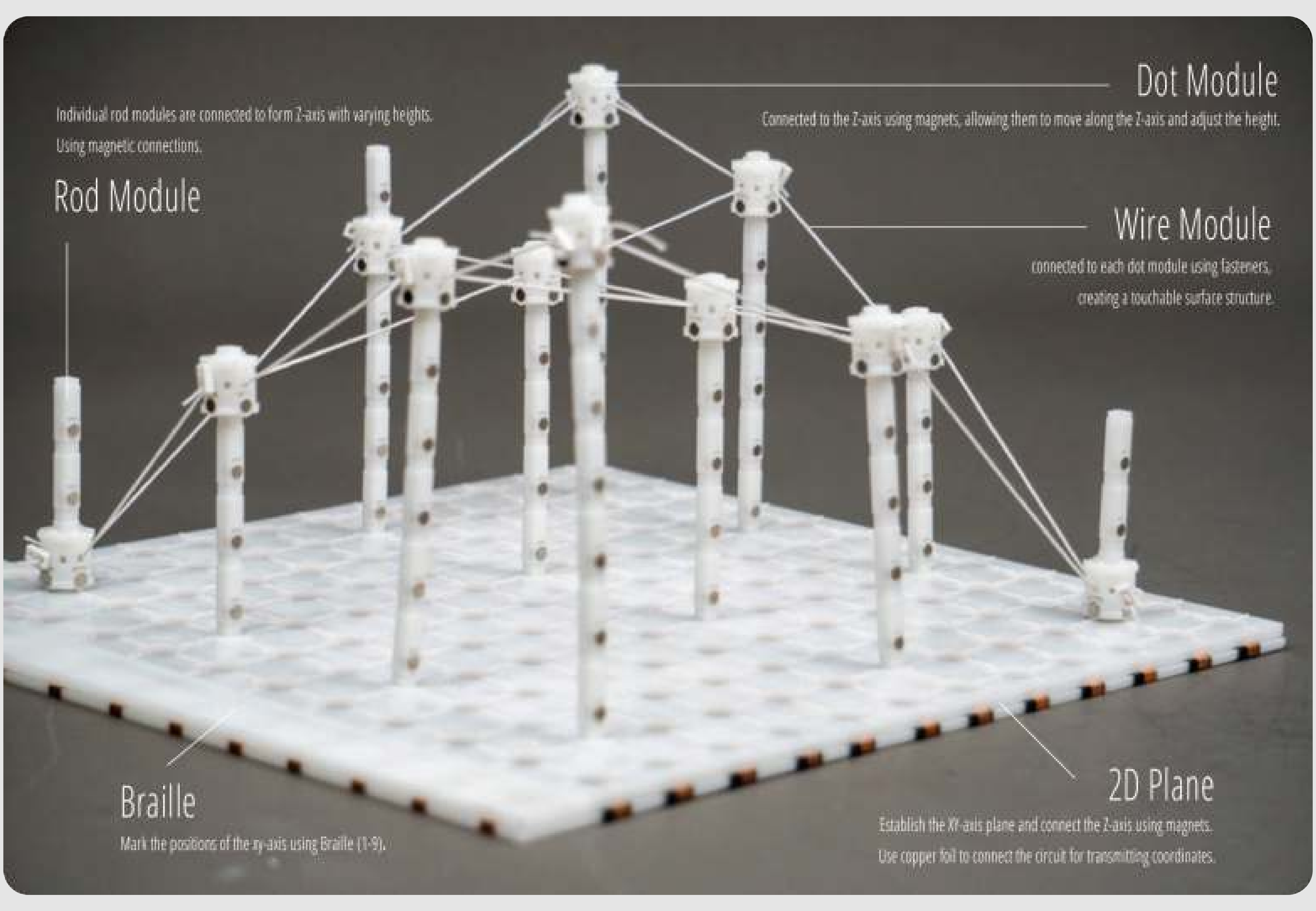
Simplifying the assembly process to prevent confusion caused by design flaws, such as entangled wire modules to quickly assemble the system for visually impaired peers.

4.Exploring digital interaction possibilities

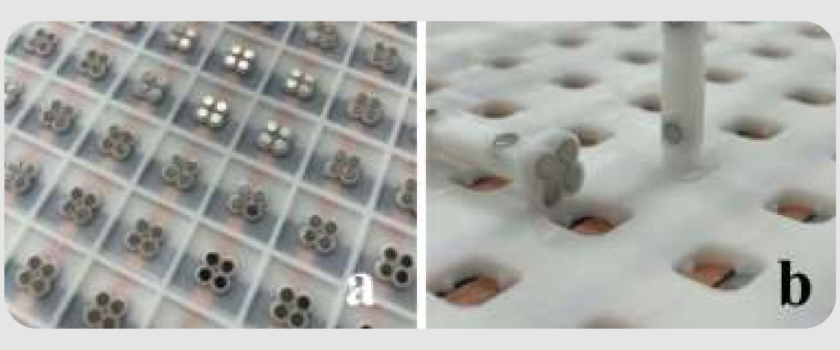
Incorporating data interfaces to transmit interactive data to digital devices, exploring potential ways to integrate digital technology into the tactile interactive system.

System Overview

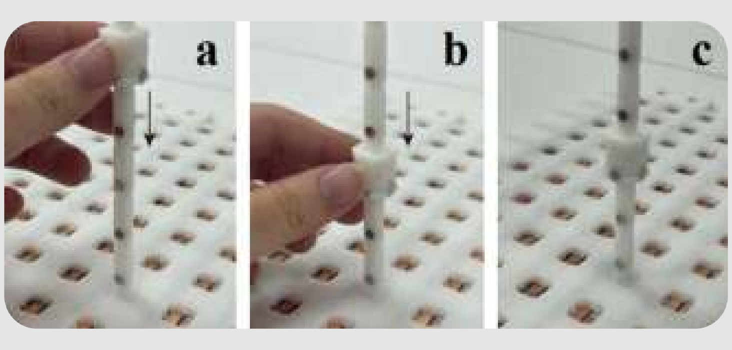
Ultimately, we have designed and created a **modular tangible interactive teaching tool**. This tool comprises **various wire modules, dot modules, and a three-dimensional coordinate system** constructed using a tablet and a supporting rod. It is based on a wireframe structure to build geometric shapes, providing a rich tactile experience of these shapes.



By optimizing the **magnetic connection design**, we enable visually impaired users to dynamically experience changes in geometric shapes and spatial relationships. Through improved plug-in structures and module design, we have gradually simplified the assembly process, allowing sighted peers to **efficiently construct models** and encouraging visually impaired users to participate as well.

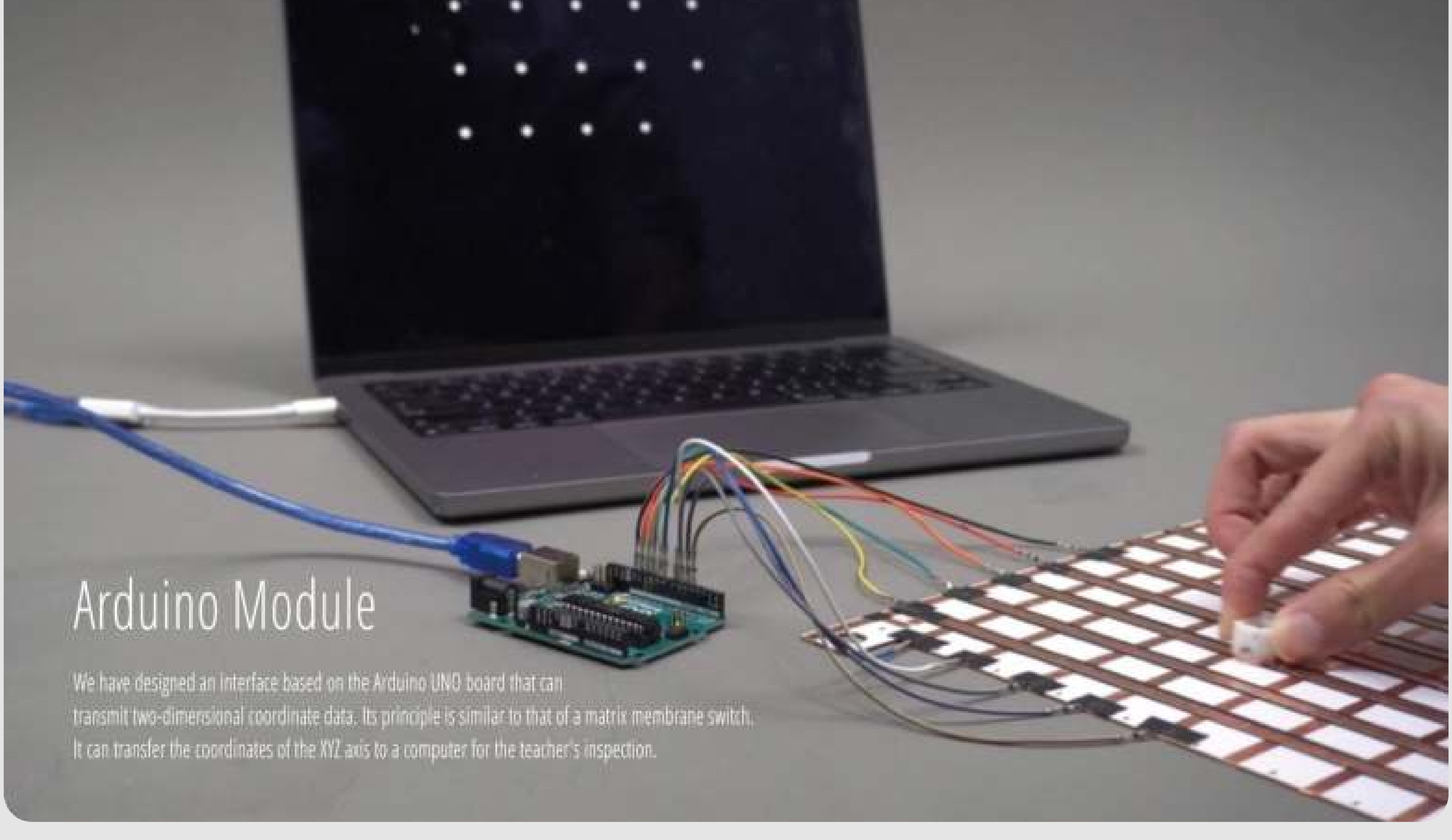


(a) Array of Magnets on the Bottom of the XY-Axis Two-Dimensional Plane; (b) Magnetic Groups on the Base of the Z-Axis Rod



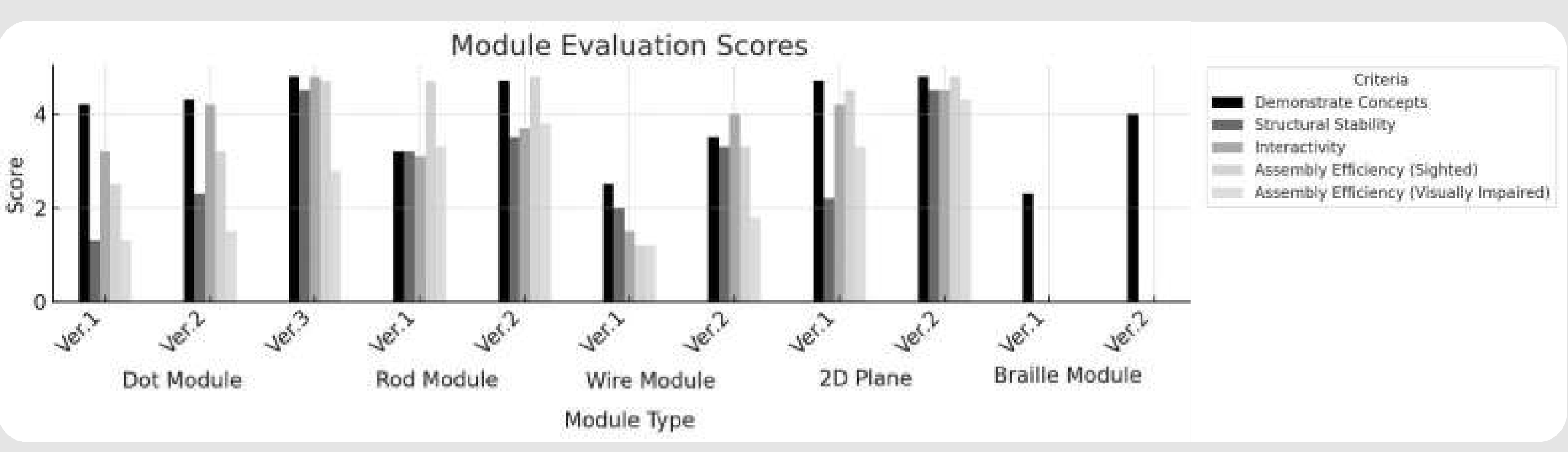
(a) Using Magnetic Attachment on the Z-Axis Rod; (b) Completion of Displacement; (c) Adhered to the Corresponding Section After Displacement

Additionally, we have also attempted to introduce **data interfaces**, aiming to enhance the design's scalability through integration with digital devices.



User Study

We conducted detailed research on the usability of each iteration version and recorded videos and text documentation. From these records, we identified multiple evaluation dimensions and organized users' subjective feedback into scores for summarization:

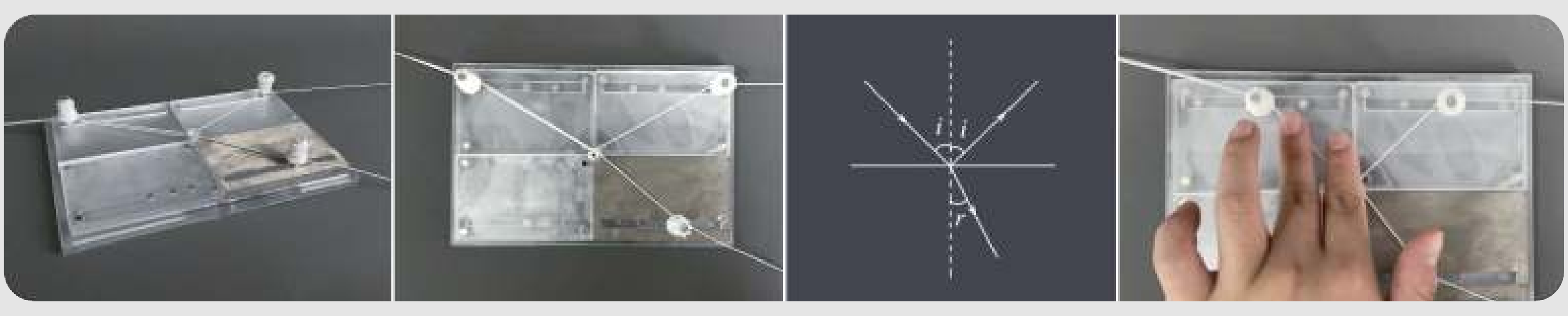


The 6 participants all completed a survey containing a Five-point Likert Scale. This result is the average of the scores from 6 participants who participated in the evaluation.

Overall, the user research and design iteration process has followed the initial design refinement approach, and there has been improvement across various evaluation dimensions. By the last iteration, **the design has achieved usability and received positive feedback from visually impaired users**, while sighted individuals can successfully assemble the system. The goal of independent assembly by visually impaired users is yet to be achieved.

#3 TactiLight

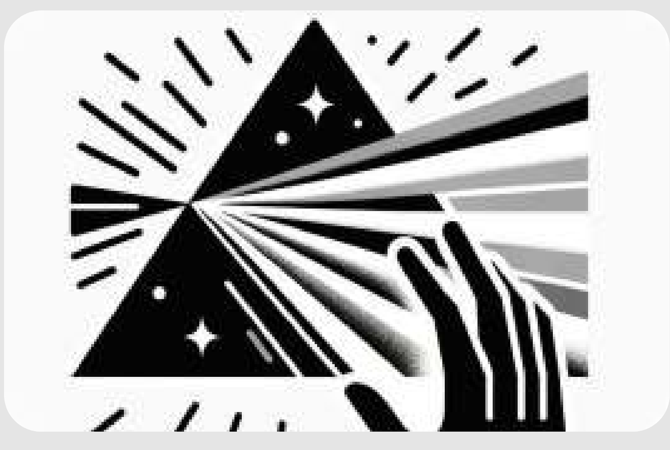
TactiLight is a modular and cost-effective tangible interaction system with various tactile feedback and auditory guidance. It allows users to explore concepts such as light reflection, and refraction by manipulating different incident materials and selecting various angles of incidence. Based on simple tests conducted with experts and visually impaired students, TactiLight has proven to be effective in aiding their understanding of optical concepts.



Background Research

Based on **research and interviews** conducted with blind school students and teachers, it has been found that visually impaired students **encounter substantial hurdles** because visualizing complex aspects like the behavior of light rays and their transformations is particularly daunting when **relying solely on tactile Braille representations**.

Semi-structured interviews with blind school students

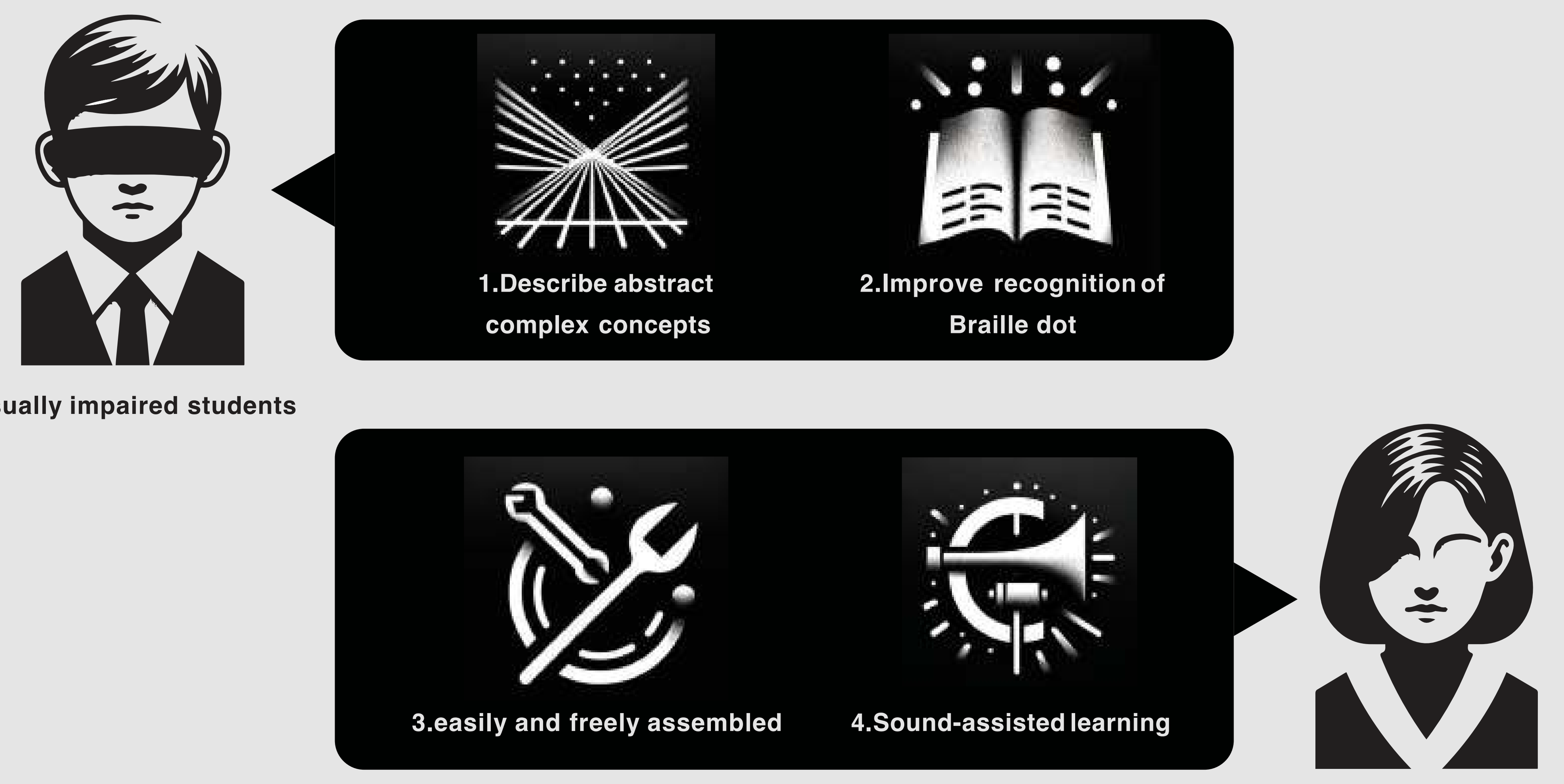


physics education

e.g. light and its various phenomena

Requirements

From semi-structured interviews with visually impaired students and blind school teachers, we identified **several requirements**:

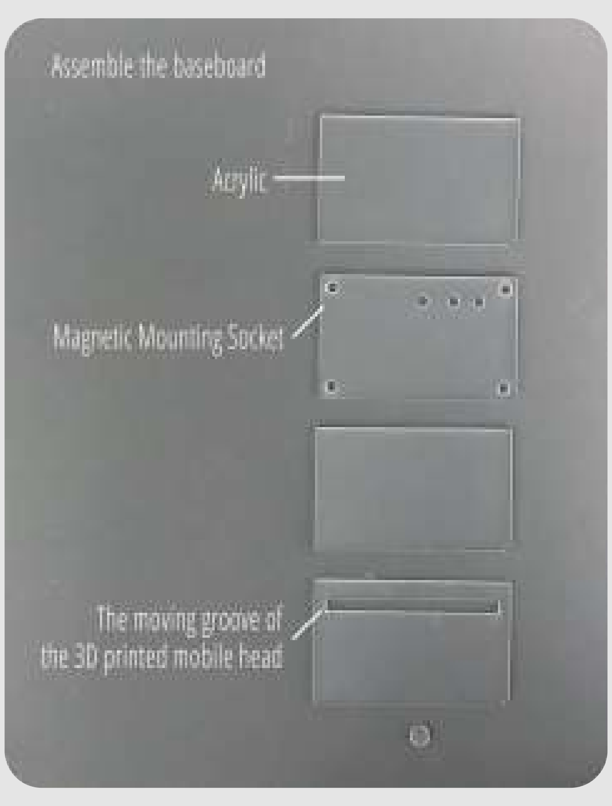


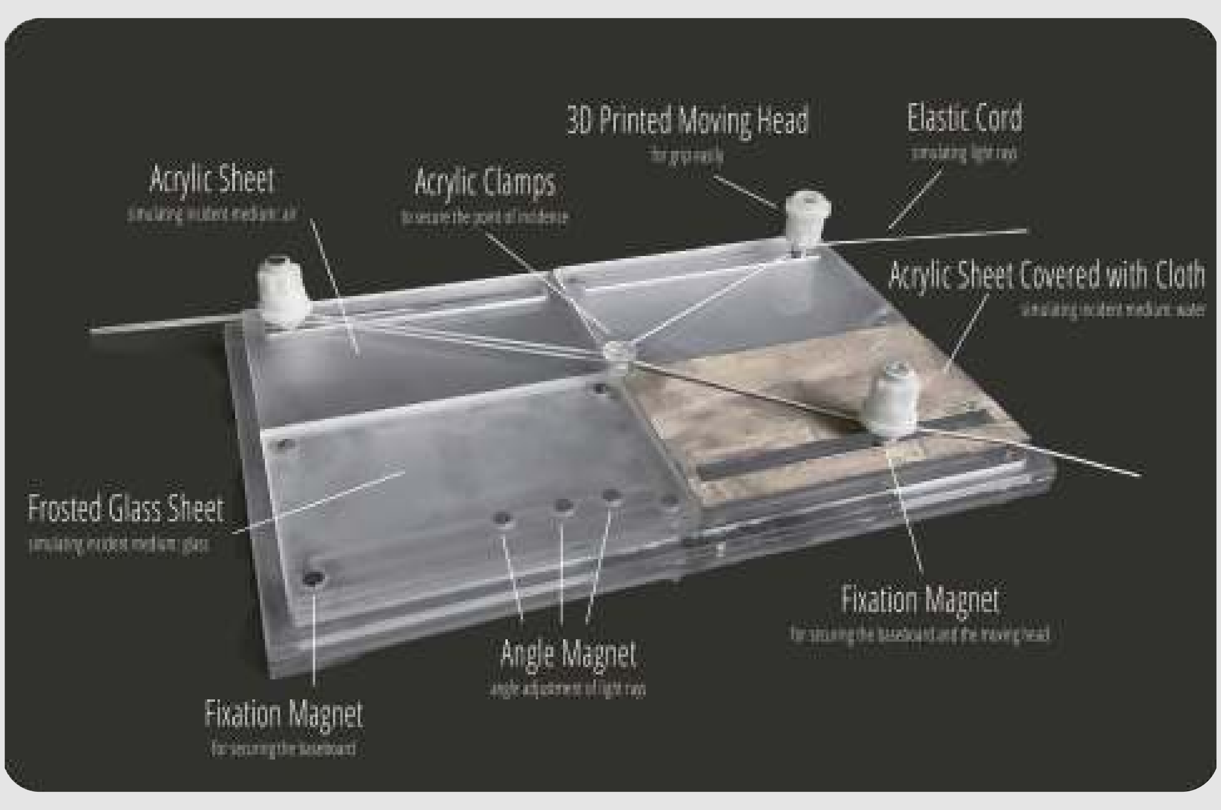
visually impaired students

blind school teachers

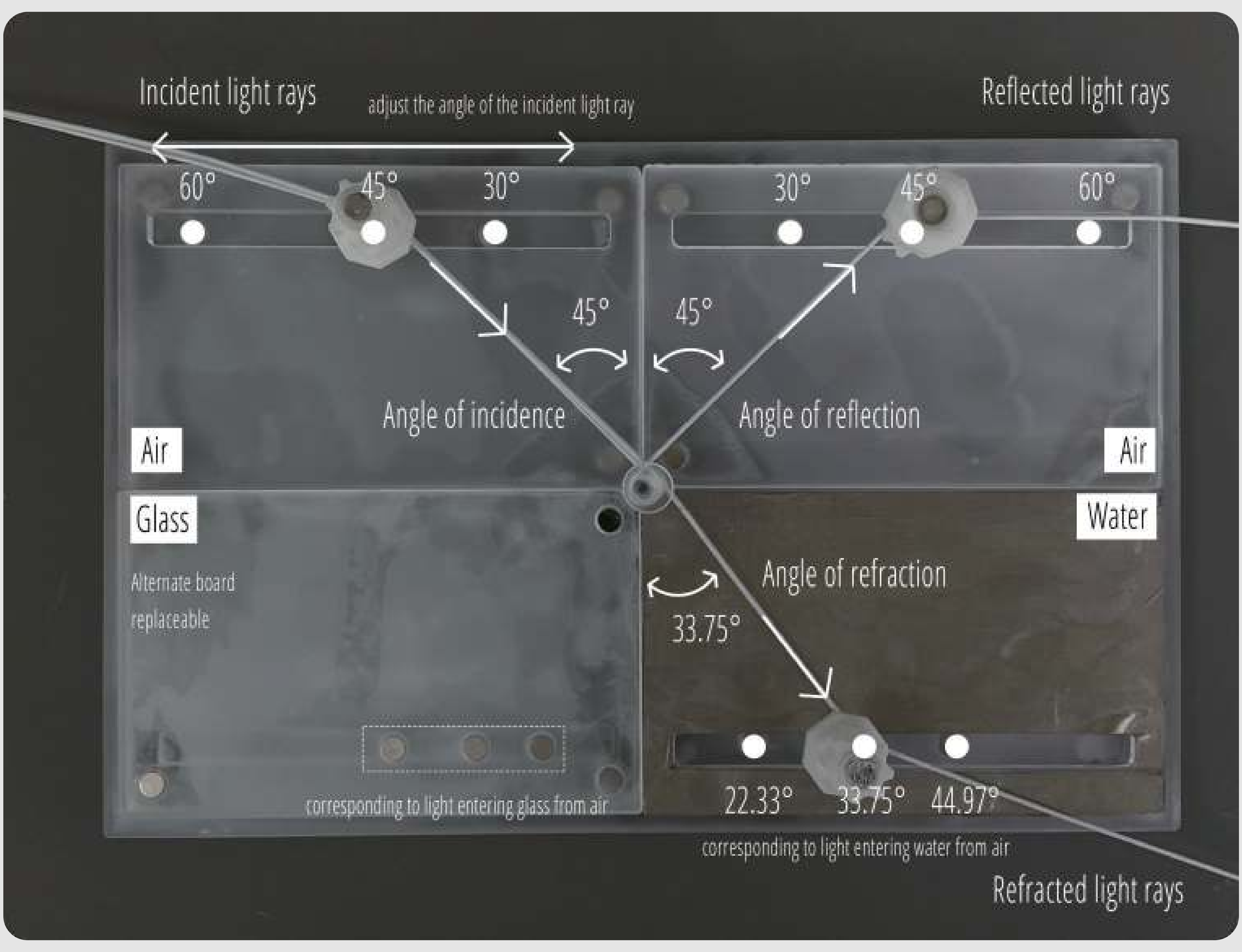
System Overview

We adopted the design line connection method from TactiGeo but referred to Myway to replace the baseboard with a **more cost-effective and lightweight acrylic board**. We use elastic cords to simulate light rays, which can be securely fastened to the **3D-printed moving head** and connected to the baseplate through **magnets**. Different elastic cords are used to simulate different light paths.



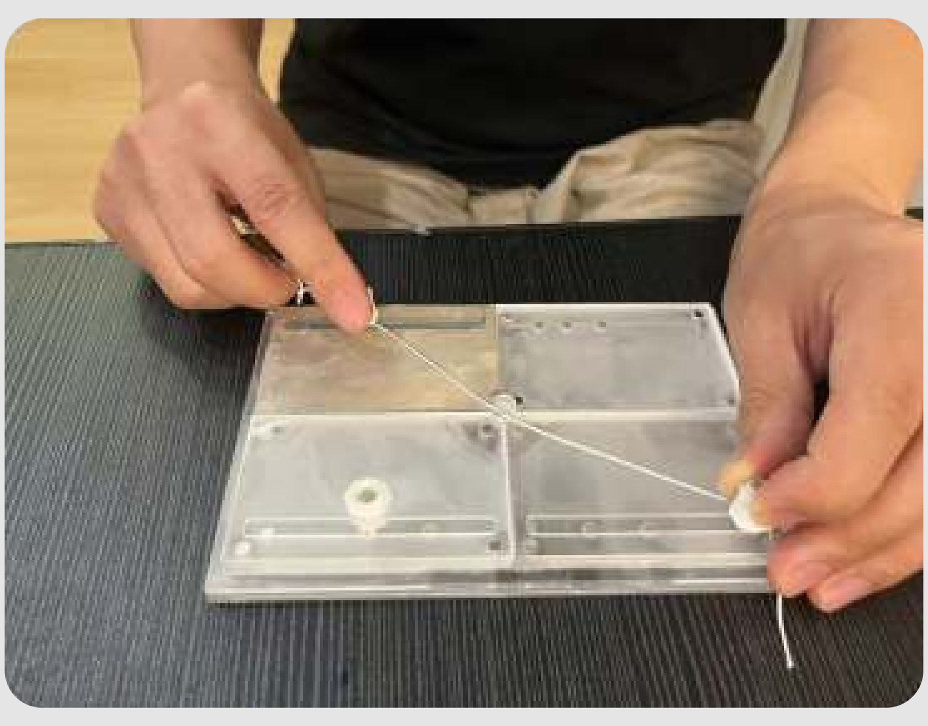


Users can adjust the angle of incidence by **moving the moving head** (e.g., adjusting it to 45°), and the magnet structure provides feedback during angle adjustments. Additionally, we employ **different materials** to simulate various mediums, including water, glass, and air, for users to choose from. Visually impaired users can determine their selected mediums by touching the different textures of these materials.



Once assembled, visually impaired users can feel the changes in light incidence, reflection, and refraction by touching the angle between the cords and the baseplate.

User Study



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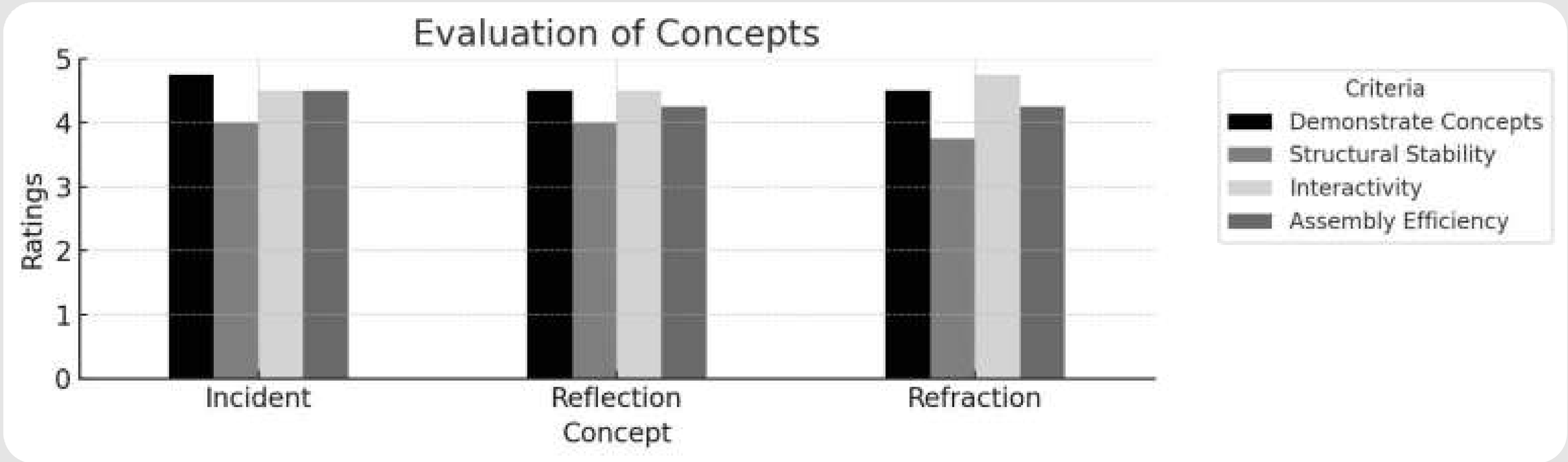
From these records, we identified multiple evaluation dimensions and organized users' subjective feedback into scores for summarization:



(The 4 participants all completed a survey containing a Five-point Likert Scale. This result is the average of the scores from 6 participants who participated in the evaluation.)

Overall, the user research and design iteration process has followed the initial design refinement approach, and there has been improvement across various evaluation dimensions. Positive feedback has been received in both **demonstrating concepts and interactivity**. Subsequently, we can focus on further enhancing structural stability to improve the **overall assembly robustness**.

#4 BrailleSticker

BrailleStickeris a 3D-printed sticker designed for low-cost modifications of accessibility devices. Visually impaired individuals can submit printing requests to designers for items, such as a touchscreen password lock, which can be parametrically printed. They can subsequently affix these Braille stickersto a wide range of objects within the environment, thereby facilitating enhanced object identification and utilization.

Background Research

Currently, with the widespread adoption of accessibilitydesign, the lives of visually impaired individuals have improved. However, **technologicaladvancements have also brought new challenges** to their daily lives.
Through research, these challenges can be primarily categorized into **three types**:

1. Use touchscreen products
e.g. a touch screen induction cooker

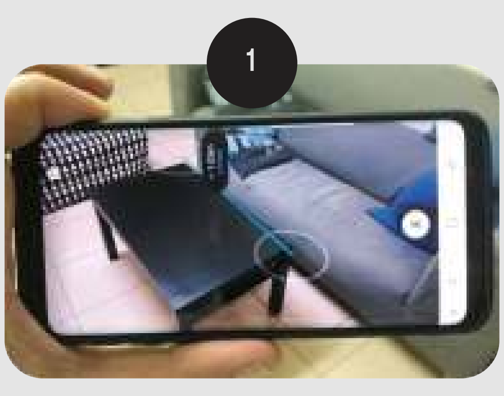
2. Distinguish similar products
e.g. personal skincare products

3. Memory complex situations
e.g. remember on which floor on now

Requirements

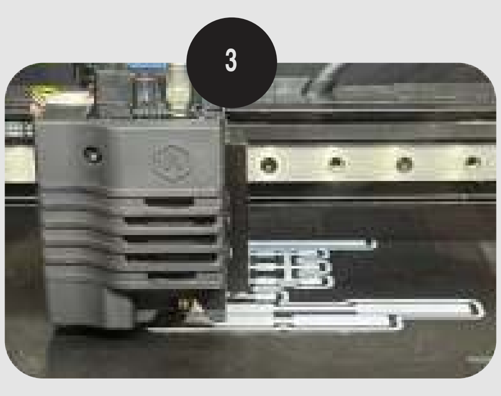
From semi-structured interviews with visually impaired individuals, we identified **several requirements**:

System Overview



Step 1: Measure
The designer visits or the visually impaired individual themselves use an AR measurement app to measure the required object's dimensions.

Step 2: Design
The designer conducts design and 3D modeling, generating Braille with Grasshopper (base height at 0.4mm and a raised height of 0.6mm).



Step 3: Manufacture
The designer employs PLA for rapid 3D printing production.

Step 4: Paste
The designer provides on-site installation assistance or sends the product to the visually impaired individual for self-installation.

How to Use

Case 1: Touchscreen Modification

We have created 3D-printed stickers with Braille based on the functions of the induction stove, facilitating use for visually impaired individuals.

We have produced 3D-printed stickers with Braille based on the size of the keypad lock, aiding visually impaired individuals in digit positioning, and using acrylic to prevent accidental touches.

Case 2: Memory-Enhancing Object Modification

We have applied 3D-printed stickers with Braille to medications enabling visually impaired individuals to touch and identify the contents of the items as needed.

We have affixed 3D-printed stickers with Braille at the turns of the staircase, allowing visually impaired individuals to touch and discern the number of floors reached when needed.

Case 3: Similar Items Modification

We have applied 3D-printed stickers with Braille to skin care products and also have created tactile symbols that can be understood by touch, so visually impaired individuals can identify the product by feeling the tactile graphic. If they can't remember, they can touch the adjacent Braille, reducing the time required for daily differentiation.

User Study

We had one visually impaired individual use the 3D-printed Braille stickers **continuouslyfor a month**. Afterward, we conducted semi-structured interviews, requesting him to provide ratings on a 5-point Likert scale for each 3D-printed Braille sticker in terms of ease of **installation efficiency, comprehensibility of content, and stability**:

Case	Installation Efficiency	Comprehensibility	Stability
Stove	4	4	4
Keypad Locks	5	5	5
Staircase	4	4	5
Medication	5	5	5
Skin Care Products	5	5	5

The results indicated that the **Braille's comprehensibility and the convenience of tactile graphics were highly satisfactory**. Furthermore, the 3D printed materials offered the advantage of integrated printing, making **installation more straightforward**, and exhibited **greater durability** in comparison to conventional paper Braille.